

Features and Image Matchⁱ

© UW CSE vision

faculty Lecture 6

Suppose
create a



you want to
pan^o



What is the first step?



Need to match portions of image^s

Solution: Match image regions using l^0





by Diva Siaⁿ

by swash^f



by s^c by Diva Siaⁿ

Harder still?



NASA Mars Rover image^s

Answer below (look for tiny colored squares



NASA Mars Rover image^S

with SIFT feature matches^S

Figure by Noah Snavely

Features can also be used for object re^c_{Feature}
Descriptor^s

Why local feature^s

Locality

features are local, so robust to occlusion and clutter^r •

Distinctiveness:

can differentiate a large database of objects •

Quantity

hundreds or thousands in a single image •

Efficiency

real-time performance achievable •

Generality

exploit different types of features in different situations •

Applications

Features are used for:

- Image alignment (e.g., panoramic mosaics)

- Object recognitionⁿ
- 3D reconstruction (e.g., stereo)
- Motion tracking^g
- Indexing and content-based retrieval^l
- Robot navigationⁿ
- ...

Want features that are invariant to tran^s.

geometric invariance: translation, rotation, scale^a •

photometric invariance: brightness, exposure,

Feature Descriptors

What about edges?

Edges can be invariant to brightness changes^b not
invariant to other transformations^s



•

What makes a good feature?

Want uniqueness^s •

- Leads to unambiguous matches in other i^m

Look for “interest points”: image regions that •

How to define “unusual”? •

Finding interest points in an image^e

Suppose we only consider a small window of pi^X . • What

defines whether a feature is a good or bad candidate Slide adapted from Darya

Frolova, Denis Simakov, Weizmann Institute

Finding interest points in an image

How does the window change when you shift it?

“corner” sign direction minimum

Find locations such that the minimum change caused by the

window in any direction is large

Slide adapted from Darya Frolova, Denis Simakov, Weizmann Institute

“edge”: “flat” region

no change along the no change in all

edge directionⁿ direction^s

Finding interest points (Feature Detection)

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e	r	b	t	of
l	e	e	e	squ
s	e	f	r	ared
i	a	o	u	diff
n	c	r	s	eren
•				ces
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	p	a	n	(SS
c				D)

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Plugging this into the formula on the previous slideⁱ Small

motion assumptionⁿ

I∴

Taylor Series expansion of

$$I(u, y + v) \approx I(x, y) + \frac{\partial I}{\partial x} u + \frac{\partial I}{\partial y} v$$

u, v is small, then first order approximation (

$$\begin{bmatrix} u \\ v \end{bmatrix} \approx \begin{bmatrix} I_x & I_y \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} + I(x, y)$$

If the

Feature detection: the math

Find these directions by looking at the eigenvectors

[

)

|

[

You can move the center of the green window to a blue unit circle

so we find directions that will result in the largest

])

|

|

|

y

]

$$I^2 y$$

Feature detection: the mat^h

x

I

I

y

$$\begin{matrix} 2 \\ x \end{matrix} I$$

$$\begin{matrix} I & & x \\ \lceil & & I \\ & | & \\ & & | \\ & & \lfloor \\ & & W \end{matrix} H$$

$$\sum^{\in} ($$

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y

$$\left(\begin{array}{c} | \\ | \end{array} \right)_x$$

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Quick eigenvalue/eigenvector review¹

eigenvectors of a matrix

x that satisfy

A are the vectors

x

λ is the **eigenvalue** corresponding to The scalar

$$Ax = \lambda x$$

The eigenvalues are found by solving:

$$\det(A - \lambda I) = 0$$

is a

$$\det \begin{bmatrix} h_{11} - \lambda & h_{12} \\ h_{21} & h_{22} - \lambda \end{bmatrix} = 0$$

matrix, so we have $\mathbf{H}$

$\mathbf{A} =$ In our case,

•

The solution: •



1

x

b

x y solving

λ , you find

Once you know

2

x

[
|
|

Capture shifts with the smallest and largest change (

)
|
|
|
|

y
|]

$$I^2 = y$$

Feature detection: the math

Eigenvalues and eigenvectors of H

I

I

$\hat{x}^+ =$ direction of **largest** increase in E .

$\hat{x}^-$

$\hat{x}^- =$ direction of **smallest** increase in E .

I

x

y

I

direction x

$$\sum_{x \in W} I_x$$

$$\begin{pmatrix} | \\ | \end{pmatrix}_x \quad] \quad E$$

$$v \quad ($$

$$u$$

$$[\approx$$

$$(,)^+$$

$$v$$

$$\cdot$$

$$x \cdot$$

$$\lambda \cdot$$

$$\cdot$$

$$u$$

$$\cdot$$

Feature detection: the math

$\mathbf{x}^-$ relevant for feature det^t₊, λ^- , and $\mathbf{x}_+$

λ How are

λ^2

λ

-

λ

$\lambda_{++} \gg \lambda$

λ_- are both

+ and λ

large;

E increases in all

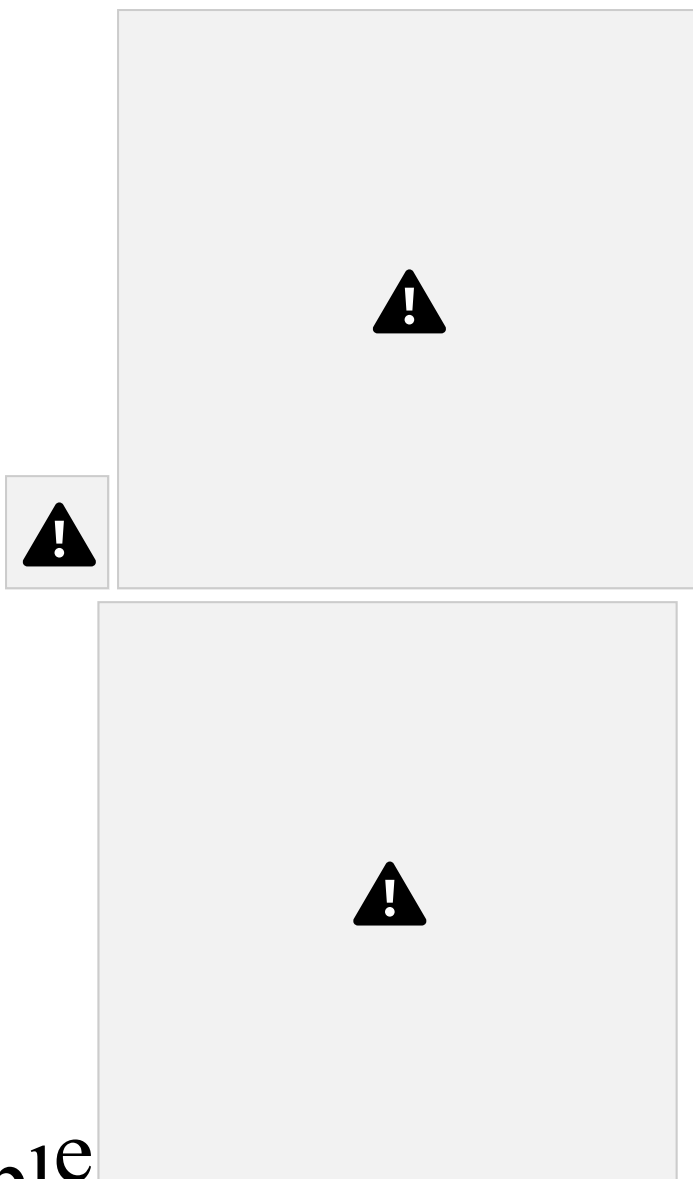
directions

-

λ

+ $\gg \lambda$

λ^1



Example^e

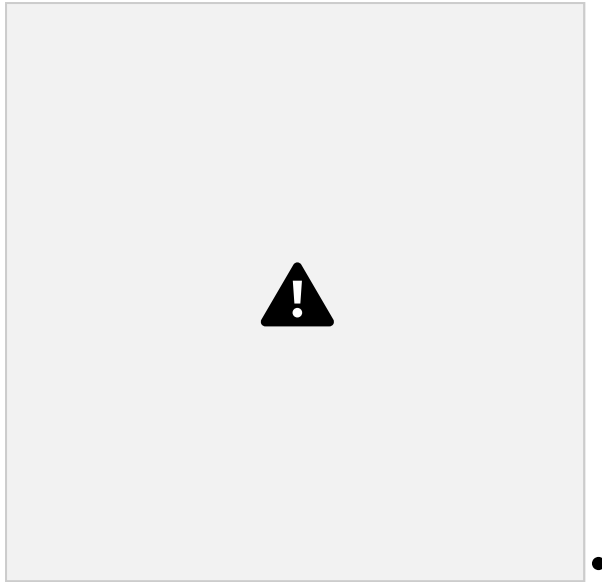


Feature detection: the math

Want $E(u, v)$ to be *large* in *all* directions

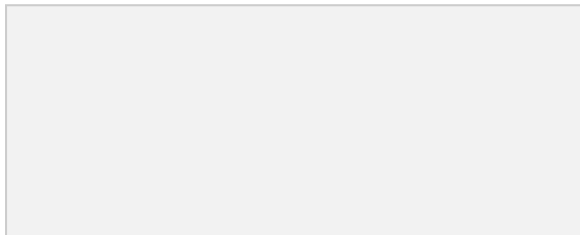
the *minimum* of $E(u, v)$ should be large over all unit •

λ^- this minimum is given by the smaller eigenvalue



-  λ Look for large values of $\bullet$

Feature detection (interest point detection) Here's



what you do

Compute the gradient at each point in the image •

matrix from the entries in the gradient H

Create the •



Compute the eigenvalues

$\lambda^- > \text{threshold}$) λ^- (i.e., Find points with large •

λ^- is a local maximum as inter^r  Choose points where •

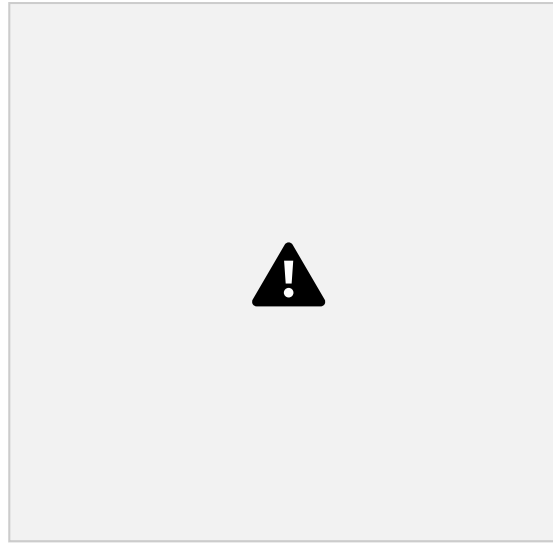
Feature detection summary

Here's what you do

• Compute the gradient at each point in the image •

• matrix from the entries in the gradient H

• Create the •



Compute the eigenvalues.

$\lambda^- > \text{threshold}$) Find points with large response (•

λ^- is a local maximum as feature Choose those points where •

- is a variant of the “Harris operator” for feature • det is the
determinant; trace = sum of diagonal ele^m • Very similar to

Called the “Harris Corner Detector” or “Harris Operator”
 k

—

)₂¹

h

₁²

h

(trac^e

k—

λ^- but less expensive (no eigenvalue
—₂²)

Harri opera^{*h*}_{*h*}

₁¹

(=

The s to^r

)₂

detector^s

Most

$H(\text{det} =$

popular

among all

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λ

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+

λ

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k

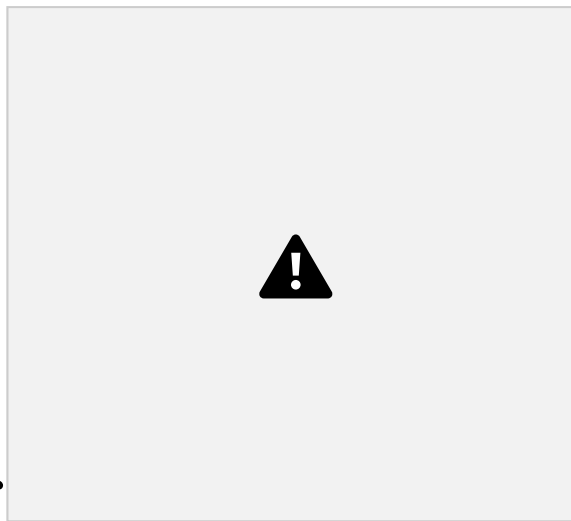
—

—

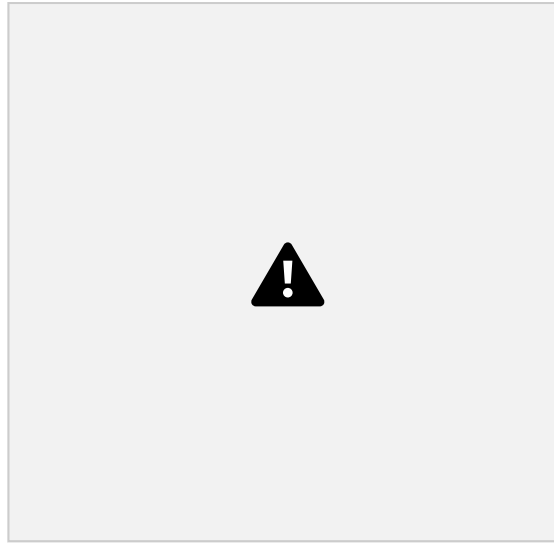
λ

λ

$$\lambda \cdot f_{arris}^H \cdot = +$$



The Harris operator



Harris detector example

f^H Harris value (red high, blue low)

Threshold ($f^H_{\text{arris}} > \text{threshold value}$)

Find local maxima of f^H_{arris}

Harris features (in red)

Invariance of Eigenvalue-based feature^e Suppose

you **rotate** the image by some angle^e

Will you still pick up the same feature points? •

Yes (since eigenvalues remain the same) •

What if you change the brightness?

Will you still pick up the same feature points? •

Mostly yes (uses gradients which involve pixel diff^e •

Scale?

No! •

Scale invariant interest point detection^C Suppose

you're looking for corners using f^H_{arris}

Key idea: Find scale that gives local maximum of f^H_{arris}

Generate copies of image at multiple scales by •

σ and convolving with Gaussians of different

using a pyramid

Find local maxima points by comparing to •

neighboring points at current and adjacent scales

Each interest point is a local maximum in both •

See SIFT paper on line for detail^S

position and scale^e

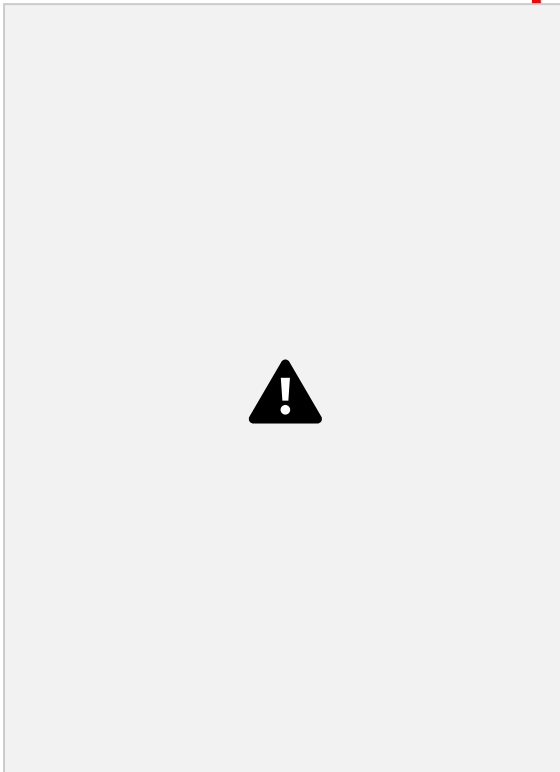
Feature descriptor^S

We know how to detect good interest point^S

Next question:

How to match image regions around interest^p **Answer:**

Need feature descriptor^s



?

Feature descriptors

Lots of possibilities (this is a popular research area)
• Simple option: match square windows of pixels
• State of the art approach: SIFT

How to achieve invariance in image matching
Two steps:

Make sure your feature *detector* is invariant 1.

Harris is invariant to translation and rotation •

Scale is trickier •

common approach is to detect features at many scales Gaussian pyramid (e.g., MOPS) –

– More sophisticated methods find “the best scale” to feature (e.g., SIFT)

2. Design an invariant feature *descriptor*

A descriptor captures the intensity information in a region the detected

feature point^t •

The simplest descriptor: a square window of pixels •

– What's this invariant to?

Let's look at some better approaches...

Rotation invariance for feature descriptor^t

Figure by

Matthew Brownⁿ

Find dominant orientation of the image patch^h

correspond^{dH}

$\mathbf{u}$, the eigenvector of $\mathbf{X}$

This is given by •

λ^+ is the *larger* eigenvalue –

Rotate the patch according to this angle •

8 pixel

Patch Oriented Multiscale

Take 40x40 square window around detected feature

Scale to 1/5 size (using prefiltering) to get 8x8 square window •

Rotate to horizontal •

Normalize the window by subtracting the mean, dividing deviation in the window •

els

pix
0
4

CSE 576: Computer Vision

Adapted from slide by Matthew Brown

Detections at multiple scales

Scale Invariant Feature Transform

Basic idea:

- Take 16x16 square window around detected interest point

- Compute edge orientation (angle of the gradient minus 90°)

- Throw out weak edges (threshold gradient magnitude)

- Create histogram of surviving edge orientations (8 bins)

angle histogram

Adapted from slide by David Lowe

SIFT descriptor
Full version
window
16x16
into a 4x4
grid of
cells
(8x8
window
below for
simplicity)
Compute
an
orientation
histogram
for each
cell
16 cells *
8
orientation
s = 128
dimensional
descriptor
•
Adapted
from slide
by David
Lowe

Properties of SIFT-based matching

Extraordinarily robust matching technique

Can handle **changes in viewpoint**^t •

– Up to about 60 degree out of plane rotationⁿ

Can handle **significant changes in illumination**: So^m day vs. night
(below) •

can r

Lots

of

code



available: •

http://people.csail.mit.edu/albert/ladypack/wiki/index.php/Known_implementation

Feature matching

¹, how to find the best match ⁱGiven a feature in I

Define distance function that compares two descriptors ⁱ₁.

2, find the one with min di^S Test all the features in I^2 .

Feature distance: SS^D

How to define the similarity between two features. Simple approach is $SSD(f^1, f^2)$

Doesn't provide a way to discard ambiguous (bad) I

sum of square differences between entries of the t^W

1

—
—

f
1

Feature distance: Ratio of SSDs

How to define the difference between two features. Better

approach: ratio distance = $SSD(f^1, f^2) / SS_{f^2}$ is best SSD match to f^1 in

I 2

2^n_d



will have ratio close to $1 -$

which have low ratio $0 -$

f
 1

I
1

Image matching

SSD feature distance^e



Suppose we use SS^D

5^0

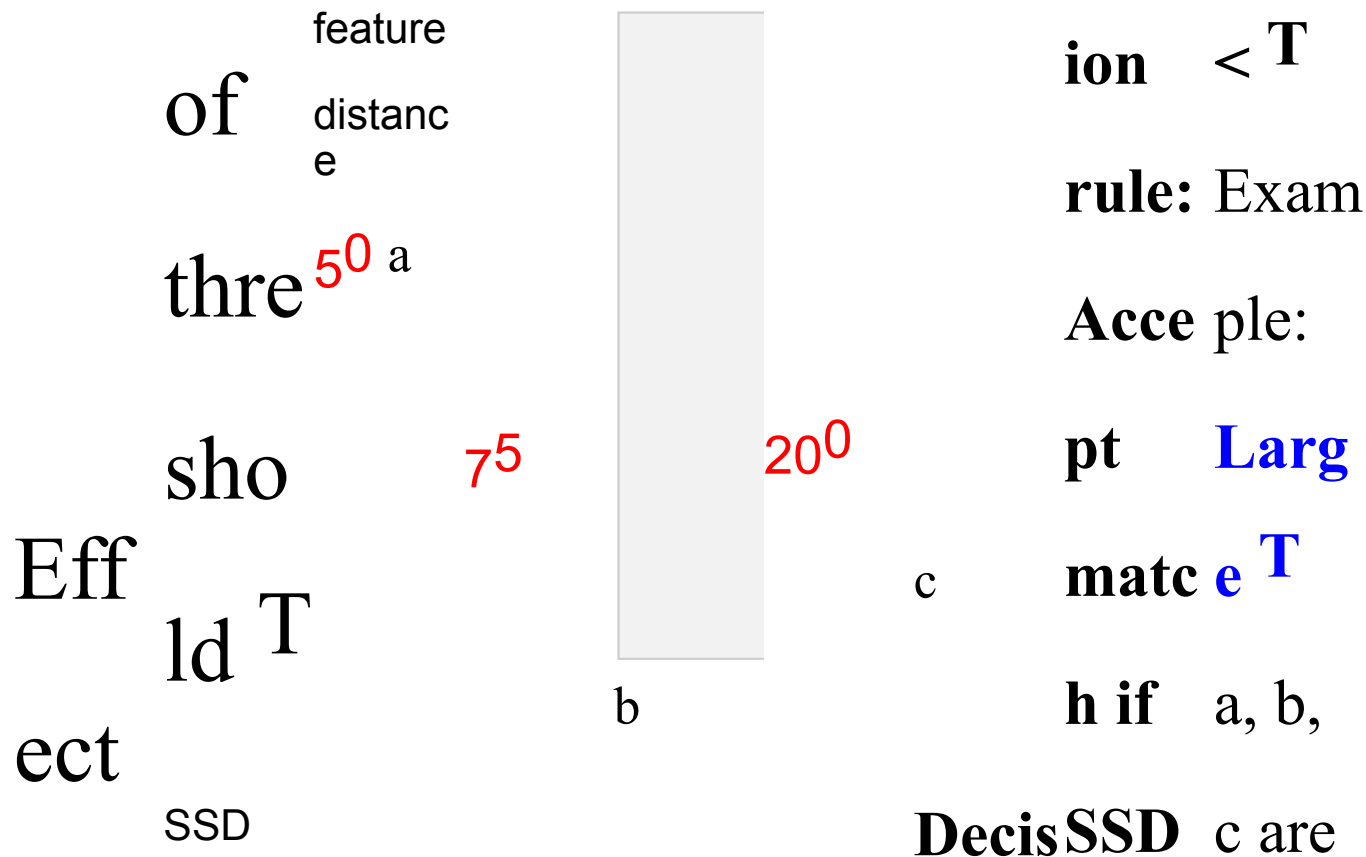
7^5

20^0

Small values are possible matches but how ^s Decision

rule: Accept match if $SSD < T$ where T is a threshold

What is the effect of choosing a particular T



all these "false positives" match false e actually
 $\Rightarrow$ a and ("true" - they e^s match positive y not a
 accepted as T = b are e are h ve") match
 250 true actually
 match positive ("false" -
 match y c is a

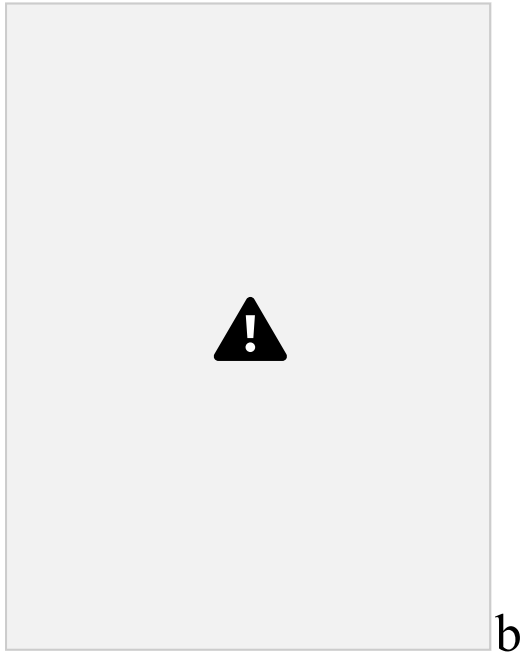
c is no longer a "false positive" (it is a "true negative")

Effect of threshold T

SSD feature distance^e

50

75



20^0

c

Decision rule: Accept match if $SSD < T$

Example: **Smaller T**

only a and b are accepted as match^h $\Rightarrow$

$T = 100$

a and b are true matches (“true positives”)

-match^s

tr^u #

ac^t #

a^c # #

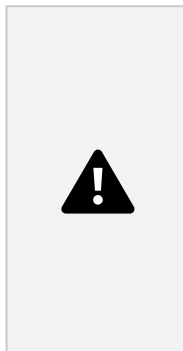
True positives and false positive^s

Actua^l Actua^l

Noⁿ Matche^s

Rejec^t

Accep^t



frequency

SS^D

T

True positive rate = True positive^S

(TPR) False positive^S

False positive rate =

(FPR)

Receiver Operating Characteristic (ROC) cur^V

Actual^I Actual^I

-matche^S Noⁿ Matche^S

SS^D

TP^R

frequency

ROC^C

1
1

FP^R

0

A^r

1

AdaP

If the features selected were bad...

FP^R

1

0

TP^R

A^r

1

AdaP

FP^R

If features selected were good...

1

0

TP^R

Using ROC curves

- Useful for comparing different feature matrices
- Pick method that maximizes area under the curve
- More info:
http://en.wikipedia.org/wiki/Receiver_operating_characteristic

0.7

true

positive

1

false positive rate

0.1

0

rate

Applications of Feature^s

- Image alignment (e.g., mosaics): Project #2
- Object recognition
- 3D reconstructionⁿ
- Motion tracking
- Indexing and database retrieval¹
- Robot navigationⁿ
- ...

Object recognition using SIFT

<http://www.cs.ubc.ca/~>

Object recognition using SIFT^T

Sony Aib⁰

SIFT usage:

Recognize 

charging

stationⁿ

Communicate^e 

with visual

cards

Teach object 

recognition

Next Time: Panoramic mosaics and images
Things to do:

- Work on Project 1

Have a

- Read online readings on mosaics

